Tradeoffs between carbon assimilation and release reveal a key role of the tropics in long-term globa-scale carbon sequestration

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The quantification of terrestrial ecosystems' carbon sequestration is crucial for understanding their role in mitigating global warming, with proper assessment requiring consideration of temporal scale. Our study quantified carbon sequestration by integrating carbon inputs and outputs over time to analyze the trade-off between high carbon assimilation rates in regions such as the tropics and low carbon losses through respiration related to high latitudes, while also assessing the effect of disturbance-related carbon losses. For the analyses, we utilized results from 9 CMIP6 Earth System Models. Then we compared the findings with the results of non-coupled models (TRENDY) and in-situ measured data-based products (CarboScope, FLUXCOM-X, CAMS). Results show that when only assimilation and respiration are considered, tropics exhibit higher carbon sequestration during 1850-2014, but the inclusion of disturbances shifts higher sequestration to high latitudes. Comparisons among products in 2000-2014 reveal similar trends between CMIP6 and FLUXCOM-X for carbon sequestration without disturbances, albeit with lower FLUXCOM-X magnitudes. When disturbances are considered, most products indicate sequestration amounts in high latitudes, except CAMS, which favors mid-latitudes. These findings underscore the significant impact of disturbances on terrestrial carbon cycles over the past century, suggesting that ecosystems, particularly in the tropics, have the potential to sequester more carbon than they currently do.

Global warming mitigation | Terrestrial ecosystems | CMIP6 | Carbon cycle

The capacity of the terrestrial biosphere to take up and store about a third of anthropogenic carbon emissions derived from the combustion of fossil fuels depend to a large degree on the long-term balance between carbon inputs (assimilation) and outputs (release), which fluctuate over time and space due to natural processes and human activities (2). Carbon inputs to terrestrial ecosystems are controlled primarily by photosynthesis, while carbon outputs arise from respiration of vegetation and soils as well as disturbances such as wildfires and land-use changes (3).

These carbon fluxes exhibit pronounced temporal variations over both short and long timescales. For example, photosynthesis typically peaks during growing seasons, while respiration rates are closely tied to temperature fluctuations. Climate change and human activities, such as wildfires and land management, can further alter these fluxes (4–6). Neglecting these temporal dynamics may lead to inaccurate estimates of an ecosystem's carbon sequestration potential (7). Spatial variability also plays a key role in ecosystem carbon dynamics. Tropical regions, with their favorable climate conditions—longer growing seasons, abundant sunlight, higher temperatures, and greater water availability—generally exhibit higher photosynthesis rates compared to extratropical regions. However, these same high temperatures and humidity in the tropics also drive elevated respiration rates by accelerating microbial activity and organic matter decomposition. In contrast, extratropical ecosystems face climatic constraints that limit gross primary production (GPP), but their cooler temperatures suppress respiration rates, leading to lower carbon outputs (8–10). As a result, while tropical ecosystems sequester more carbon in absolute terms, extratropical ecosystems may retain a higher proportion of the carbon they assimilate due to their reduced respiration losses. The balance between carbon inputs and outputs is further complicated by disturbances such as wildfires and land-use changes, which can drastically alter carbon fluxes (1, 6). This raises an important question: Is carbon sequestration primarily driven by the high assimilation rates in the tropics or by the lower respiration rates in the extratropics? Moreover, how do disturbances influence this balance?

These questions should be addressed by a meas accounts not only for carbon stocks in the ecosyst also for how long that carbon remains stored. T carbon remains stored is one of the biggest uncertaint projecting how terrestrial vegetation will respond to i atmospheric CO_2 (11). Consequently, we use the Sequestration (CS) metric. CS is calculated as the ar the curve of remaining carbon over a selected time resulting from the balance between carbon inputs and (12). The units of CS are mass multiplied by time (e · year). This metric involves two integral calculation effectively capture how long ecosystems keep carbo the atmosphere (see Annex X) and distinguish itself so-called tonne-years methods.

We calculated CS for the period 1850–201 monthly outputs from gross primary productivity heterotrophic respiration (r_h), autotrophic respirat and net biome production (NBP), based on da nine Earth System Models (ESMs) participating Coupled Model Intercomparison Project Phase 6 (These models explicitly track carbon fluxes across t system. Additionally, we calculated CS for a shorte (2001–2014) using results from five different produ model-based datasets (CMIP6 and TRENDY) ar products derived from in-situ measurements (Carl FLUXCOM-X, and CAMS).

By accurately quantifying terrestrial carbon seque we aim to deepen our understanding of ecosystem dynamics to inform better land management pract enhance strategies to maximize ecosystems' cap mitigate climate change effectively.

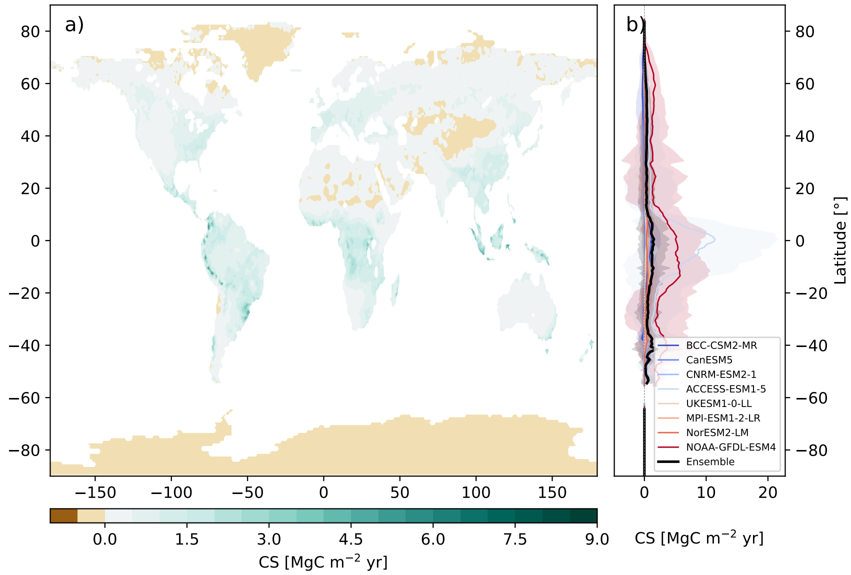

Fig. 1. Carbon Sequestration (CS) spatial patterns from NEP from CMIP6 models (1850-2014). a) Global distributions of CS based on the weighted ensemble of 8 CMIP6 models, and b) latitudinal CS profiles for individual CMIP6 models (colored lines) and their ensemble (black line). The dotted line represents zero carbon sequestration and the shadow areas represent two standard deviations from the mean.

Results

Spatial variations of Carbon Sequestration. Carbon Sequestration (CS) calculations using NEP from eight CMIP6 models (Supplementary Table 1 (see SI.tex)) reveal a predominantly positive global distribution, with notable exceptions in polar regions and desertic zones such as Sahara, Atacama, Arabian, Central Asia and Gobi deserts (Fig. 1a). Forests exhibit the highest CS values, particularly in the Amazon, Eastern USA, Congo basin, Western Europe, Indonesia, Borneo, New Guinea, and Eastern China. The latitudinal CS mean shown in Fig. 1b demonstrates positive CS across all latitudes, indicating long-term carbon input surpasses output when disturbances are not considered. Peak CS occurs in the tropics (approximately 25°S to 20°N), while mid-latitudes (approximately 20-40°N and 20-30°S) show the lower values.

The cumulative CS across latitudinal zones exhibits a consistent trend over time in all models, albeit with varying magnitudes. NOAA-GFDL-ESM4 and ACCESS-ESM1-5 models show the highest sequestration values, while CanESM5 and BCC-CSM2-MR models show the lowest (Fig. 2a). Tropical regions have sequestered the largest amount of carbon from 1850 to 2014, followed by mid-latitudes (Fig. 2b-d). Although high latitudes sequester on average more carbon per unit area than mid-latitudes (Fig. 1b), the larger territorial expanse of mid-latitudes results in a greater total CS (Fig. 2). All CMIP6 models indicate a positive CS in the tropics during the studied period (Fig. 2b). However, in mid- and high-latitudes, at least one model (BCC-CSM2-MR) shows negative CS, where accumulated respiration exceeds GPP (Fig. 2c,d). Additionally, three models indicate nearly zero CS in high latitudes (Fig. 2d).

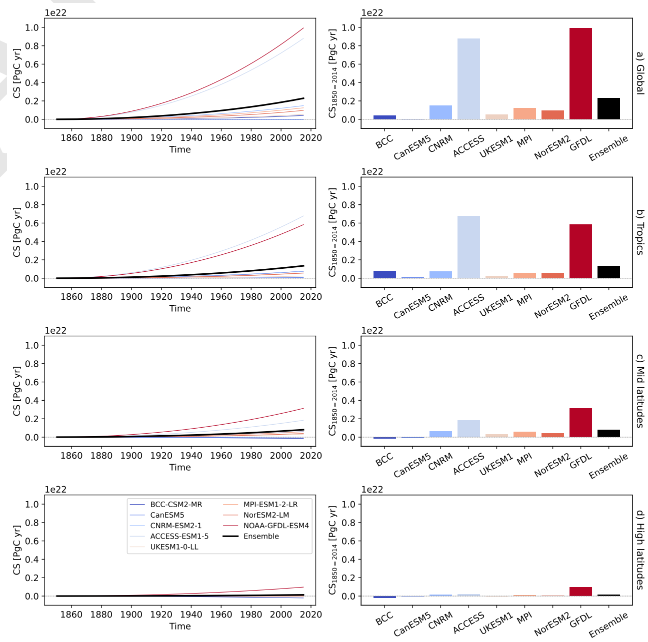

Fig. 2. Time accumulation of Carbon Sequestration (CS) (left panel) and CS at the end of the period from 1850 to 2014 (right panel) calculated using NEP for a) global, b) tropics, c) mid-latitudes, and d) high-latitudes zones calculated using the results of 8 models from CMIP6. Black lines and bars indicate the results using the ensemble of the models. The dotted lines indicate null values of CS. To save space, only the initial part of each model name is displayed on the horizontal axes in the right panel.

Disturbance impacts on Carbon Sequestration. The previous results, based on NEP, only account for climate variations and exclude disturbances like deforestation and fires. When using Net Biome Production (NBP) to calculate CS instead, CS dynamics change significantly. The values of CS calculated with NBP are about two orders of magnitude smaller than with NEP (Figs. 2a and 3a) and are predominantly negative.

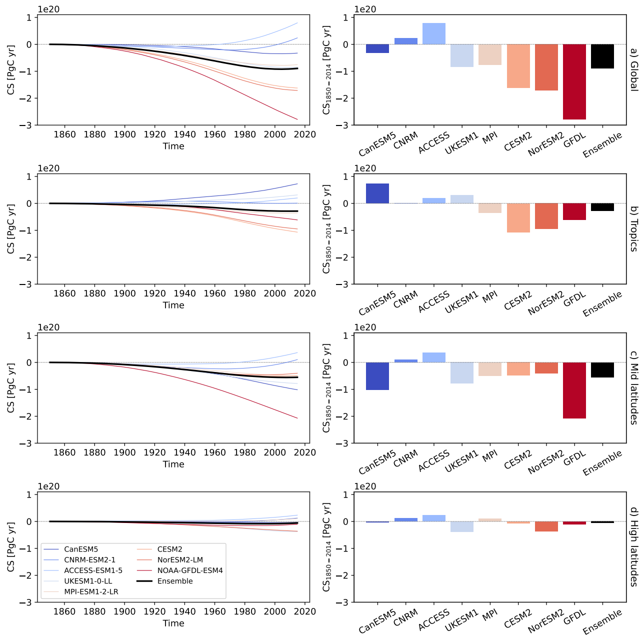

Fig. 3. Time accumulation of Carbon Sequestration (CS) (left panel) and CS at the end of the period from 1850 to 2014 (right panel) calculated using NBP for a) global, b) tropics, c) mid-latitudes, and d) high-latitudes zones calculated using the results of 8 models from CMIP6. Black lines and bars indicate the results using the ensemble of the models. The dotted lines indicate null values of CS. To save space, only the initial part of each model name is displayed on the horizontal axes in the right panel.

Globally accumulated CS is negative for all CMIP6 models except CNRM-ESM2-1 and ACCESS-ESM1-5, with the ensemble indicating a carbon debt of approximately 0.8 PgC yr from 1850-2014 (Fig. 3a). This indicates that terrestrial ecosystems lost more carbon than they gained during this period. High latitudes show the least negative values, followed by the tropics (Fig. 3b-d). Negative CS values are observed in 5 out of 8 models for the tropics, 6 for mid-latitudes, and 5 for high latitudes (Fig. 3b-d). Notably, the ensemble accumulation of CS over time is no longer monotonic as when using NEP; after decreasing from 1850 to around 1970, it slightly increases.

Comparison of CMIP6 carbon sequestration with other data- and model-based products. Carbon Sequestration calculations for 2001-2014 consistently show higher values when derived from NEP compared to NBP (left and right panel of Fig. 4, respectively). NEP-based CS from CMIP6, TRENDY, and FLUXCOM-X indicate that tropical regions have the highest values, while high latitudes have the lowest, aligning with previous results in Fig. 2. CMIP6 ensemble yields the highest global CS values, while TRENDY shows the lowest. Although both ensemble models display similar latitudinal zone trends, CMIP6 consistently surpasses TRENDY by up to 250 PgC globally. This substantial difference observed despite some TRENDY models being used in CMIP6 Earth System Models underscores the impact of coupling terrestrial models with climate models and the explicit modeling of carbon movement throughout the Earth System in CMIP6. Then, when using NBP, mid-latitudes show the highest CS values for CMIP6, TRENDY, and CAMS, but not for CarboScope (left panel in Fig. 4). This finding is consistent with the

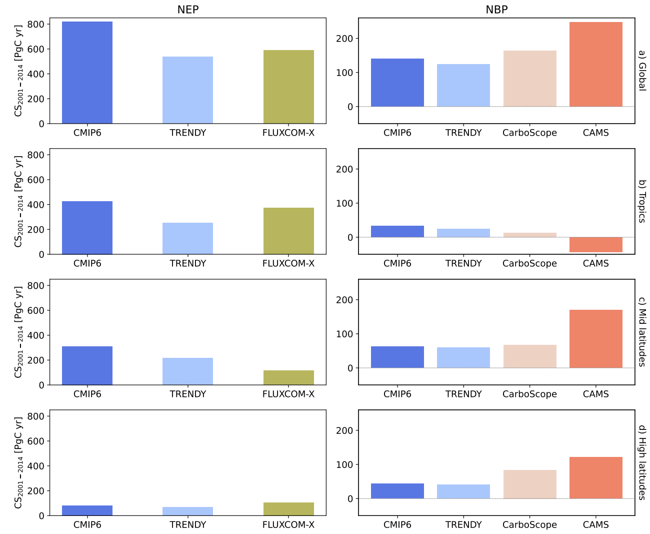

Fig. 4. Carbon sequestration (CS) across different products (2001-2014) calculated using NEP or NEE (left panel) and NBP (right panel) for a) global, b) tropics, c) mid-latitudes, and d) high-latitudes. The dotted lines indicate the null values of CS. CMIP6 and TRENDY CS values correspond to the ensemble of their respective models. Note that the y-axis of the left and right panels do not have the same scale.

CMIP6 results for the 1850-2014 period. However, when examining the common 14-year timeframe (2001-2014) across all products, CS values are predominantly positive, except for CAMS in tropical regions. This shift towards positive values likely stems from the increasing trend in CS observed in Fig. 3 from around 1970 onwards. In this shorter timeframe, tropical regions shift from having the second-highest CS to the lowest, with CAMS even indicating a negative value. For NBP-based calculations, data-driven models (CAMS and CarboScope) show higher CS values than the model-based products. CMIP6 and TRENDY yield similar CS values when using NBP, with CMIP6 slightly higher, but the difference is less pronounced compared to NEP-based calculations.

Discussion

Trade-off high carbon assimilation and low respiration. Terrestrial ecosystems worldwide have absorbed more carbon than released from 1850 to 2014, acting as net carbon sinks when disturbances are not considered. This is evident from the positive CS values across all latitudes (Fig. 1b) and the steady increase in the left panel in Fig. 2. These findings align with the growing global carbon budget and previous research that has observed increases in NEP over various periods and using different methods and databases (1, 13–15). Several factors explain this trend, including increased atmospheric CO_2, higher nitrogen deposition, and the effects of climate change. These factors enhance photosynthesis via fertilization, improve plant water use through reductions in stomatal conductance in water-limited ecosystems, reduce cold limitations at high latitudes, and extend growing seasons (1, 13, 16–18). While elevated CO_2 can positively and negatively affect carbon sequestration, the overall impact up to 2014 appears positive (right panel in Fig. 2). The trends remain consistent even when focusing solely on the 2001-2014 period (Left panel in Fig. 4).

The high CS observed in forests during the studied period (Fig. 1a) supports earlier research indicating that forests, particularly tropical and boreal regions, are the primary reservoirs of the world's biomass carbon (19). This is likely because forests are dominated by trees with large woody masses, and most of the carbon captured by vegetation is allocated to woody tissues (20). Their complex structure also makes forests highly efficient at capturing carbon, and when vegetation dies, it contributes to soil organic carbon, further enhancing storage (21). Over time, increases in NEP have been observed in forests, especially in those without nutrient limitations (13, 16, 22).

The tropics show higher carbon sequestration values when data are divided by latitudinal zones (Figs. 1b and 2b), indicating that high tropical assimilation outweighs low respiration in extratropical regions when only climate is considered. This can be attributed to the dense vegetation, extensive biomass, and rapid growth in tropical areas, which contribute significantly to NEP trends (1, 9, 13, 19, 23), even with rising temperatures and lower simulated soil carbon stocks and residence times (24, 25). High and stable temperatures in these regions promote increased carbon allocation in the canopy (20). However, it's important to note that CMIP6 models don't capture potential declines in tropical forest carbon sink strength due to saturation and drought, as suggested by some studies (9, 18, 26). In contrast, high latitudes (>50°) exhibit lower CS values, likely due to temperature limitations, although they show enhanced seasonal CO2 cycles with larger summer uptakes and winter releases (27). Despite being well-studied, carbon sinks in high-latitude regions still show discrepancies among different products, and soil carbon is often underestimated, suggesting a potentially greater carbon sequestration capacity than currently simulated (18, 28).

Consequences of disturbances. When factors beyond climate, such as harvesting, deforestation, and fire, are considered, CS dynamics shift significantly. From 1850 to 2014, most models indicate that cumulative respiration exceeded assimilation 3, making terrestrial ecosystems a carbon source rather than a sink. While the Amazon, Congo, and Papua New Guinea forests still exhibit the highest CS values (Supplementary Fig. 5), high-latitude regions show the smallest carbon losses (Fig. 3). These dynamics are driven by deforestation, degradation, and drought, especially in the Amazon, as well as the negative effects of climate variability and change (9, 18, 19, 26, 29). Although African tropical forests generally show net carbon gain and intact tropical Americas continue to gain carbon (9, 18, 19) (though the sink is declining (30)), northern boreal forests seem to be benefit from climate change and accumulating carbon at higher rates than in the past (1, 9, 19, 31).

When only data from 2001-2014 is considered, CS calculated with NBP becomes positive, albeit lower than CS calculated with NEP (Fig. 4). This trend change could be attributed to increased CO_2 fertilization, longer growing seasons, enhanced nutrient cycling and nitrogen deposition, and reforestation efforts (18, 32). The forest transition in the 19th and 20th centuries, followed by forest regrowth in Europe (18), along with land abandonment, preservation, reforestation, and improved land management practices during the second half of the 20th century (33), and the acceleration in the increase rate in global temperature in the 1970s (9), likely explain the increased trend in CS observed since around the 1970s (Left panel in Fig. 3).

Accounting for disturbances, the tropics sequestered the least amount of carbon among latitude zones during 2001-2014 (Right panel in Fig. 4), with CAMS even indicating a net carbon release. This aligns with previous findings indicating high cumulative land-use emissions in the tropics and that warm climates decrease NBP due to temperature and soil moisture sensitivities (18, 34, 35). It's also important to acknowledge that while CMIP6 models capture carbon gains from tree growth, they don't accurately represent carbon losses from tree mortality (26).

Agreement and discrepancies among products. Carbon Sequestration (CS) values vary significantly among CMIP6 models and different products, as previously noted by researchers using various carbon metrics (15, 28, 31**?**). The behavior of CMIP6 models is influenced by their response to CO_2, temperature perturbations, resolutions, residence times, land surface schemes, and the inclusion of components such as nutrient cycles, fires, and dynamic vegetation (24, 31**? ?**). Models that couple the nitrogen (N) cycle are expected to reduce CO_2 fertilization and carbon releases, show a lower negative impact of global warming, and align better with observations (**? ?**). In a different CMIP6 experiment (1pctCO2), Arora et al. (31) found that CNRM-ESM2-1 and CanESM5 have the largest land carbon uptake, which only slightly aligns with our CS-NBP results (Fig. 3). Melnikova et al. (**?**) also identified CNRM-ESM2-1 as the model with the highest land carbon uptake. These models are not coupled with the N cycle, leading to fewer limitations on photosynthesis. CanESM5 exhibits a strong CO_2 fertilization effect due to retuned photosynthesis parameters and high carbon use efficiency (**?**), while CNRM-ESM2-1 shows high land carbon uptake attributed to its long carbon residence time in soils and vegetation (31). Arora et al. (31) also found ACCESS to have the lowest land carbon uptake, citing weak CO_2 fertilization due to nutrient limitations from N and P coupling. However, our study shows ACCESS as one of the models with the highest CS values, being one of the few models indicating positive CS-NBP, and CNRM-ESM2 and CanESM5 do not demonstrate notably high CS values (Figs. 2 and 3). These differences likely arise from the definition of CS used in this study. Unlike approaches that focus solely on carbon stocks, this metric considers the long-term carbon input-output balance (12), accumulating effects over time. For example, CS accumulates both the effects of low carbon uptake rates caused by N limitations and the small carbon releases due to additional N release from dead organic matter triggered by global temperature increases (**? ?**), which could explain the high CS values observed in the ACCESS model.

Our analysis did not reveal a clear pattern between models considering N cycle coupling (ACCESS-ESM1-5, CESM2, MPI-ESM1-2-LR, NorESM2-LM, UKESM1-0-L) and dynamic vegetation (NOAA-GFDL-ESM4, CESM2, and UKESM1-0-LL) and CS values. However, we observed a clear pattern of lower CS-NBP values in models incorporating fire components (NOAA-GFDL-ESM4, NorESM2-LM, CESM2, MPI-ESM1-2-LR) (Fig. 3), except for CNRM-ESM2-1. Notably, the GFDL model shifts from having the highest CS value for NEP to the lowest for NBP (Figs. 2 and 3).

This suggests that the immediate carbon losses from fires outweigh the potential long-term benefits of low-intensity fires in promoting forest carbon storage (**?**), or that these long-term effects are not considered in the models. Furthermore, the differences observed between CESM2 and NorESM2-LM results, despite both using the same land model (CLM5), emphasize the complex interactions within the entire Earth system simulation and the intricate nature of Earth system modeling, where even slight differences in parameterizations or representations of processes can lead to divergent results (37). The complexity of ESMs, which aim to simulate all relevant physical, chemical, and biological processes, means that uncertainties persist, likely due to models still missing key processes or inadequately representing certain interactions within the Earth system (28**?**).

Comparing results from different products for 2001-2014, CMIP6, TRENDY, and FLUXCOM-X agree that CS with no disturbances is highest in the tropics, followed by mid-latitudes (left panel in Fig. 4). This pattern aligns with findings from the full CMIP6 time series (Fig. 2). CMIP6 ensemble shows the highest global CS values when calculated with NEP, while the TRENDY ensemble shows the lowest as previously noted for carbon land sink (**?**). For CS calculated with NBP, CMIP6 remains higher than TRENDY, but the gap narrows (Fig. 4). The main difference between CMIP6 and TRENDY lies in their nature: CMIP6 is an intercomparison of Earth System Models (ESMs), while TRENDY compares Dynamic Global Vegetation Models (DGVMs). ESMs assimilate physical atmospheric and oceanic data, reproducing historical CO2 growth rates and carbon fluxes between atmosphere, sea, and land (18**?**). However, these products are not entirely independent, as some CMIP6 models use TRENDY as land models.

When comparing TRENDY and CMIP6 with data-driven products, it's crucial to recognize that all these approaches have inherent uncertainties stemming from various sources such as modeling assumptions and design choices, forcing datasets, representation of processes, under-sampling in biomes such as tropics, etc. (1). The FLUXCOM initiative aims to address some of these issues by creating ensemble products and performing cross-consistency checks with independent approaches (23). Our results indicate that inversion products (CAMS and CarboScope) show higher global CS-NBP values than model-based products, except in the tropics. Notably, CAMS even yields negative CS values for the tropics (right panel in Fig. 4). Explanations for differences between model-based and inversion products include: inadequate parameterization of land management and degradation (13), shorter modeled carbon ecosystem residence time (**?**), insufficient capture of volcanic eruptions effects (**?**), underestimation of northern latitude soil carbon stocks (28), poor reproduction of forests stocks (**?**) and lack of climatic and CO_2 interactions (13) by model-based products. Despite using weights based on the models' proximity to FLUXCOM-X (for NEP) and CarboScope (for NBP) when calculating the CMIP6 and TRENDY ensembles, the results from these two model groups still show substantial differences.

In the tropics, higher CS-NBP values in CMIP6 and TRENDY compared to inversion products (Fig. 4b) might be due to models not correctly balancing CO2 fertilization effects and respiration costs in Amazon, inaccurate representation of mortality trends in tropical forests (26), overestimation of biomass in tropical forests (**?**), a general tendency to overestimate carbon in tropical regions while underestimating it in the Northern Hemisphere (**?**), and an underestimation of drought impacts on tropical forest ecosystems (9).

Differences between CAMS and CarboScope inversions may stem from varying interhemispheric transport models or inversion settings (13), though the exact reasons remain unclear (35). While this study didn't directly compare FLUXCOM-X with inversion models, previous research found agreements between FLUXCOM-X and CarboScope for 2015-2020 (23). FLUXCOM-X shows higher CS values for high latitudes compared to TRENDY and CMIP6, possibly due to its incorporation of more observational data, which may better capture complex processes in northern extra-tropics (18).

Implications for climate change mitigation.

- Importance of temporal scale consideration. The research underscores the importance of considering both carbon inputs and outputs, as well as disturbances when developing climate mitigation strategies involving terrestrial ecosystems.
- Disturbances have had a significant impact on terrestrial carbon cycles over the past century, particularly in tropical regions.
- Tropical ecosystems have the potential to sequester more carbon than they currently do, suggesting opportunities for enhanced mitigation efforts in these areas.
- Different carbon cycle models and data products show varying results, highlighting the need for continued refinement and comparison of these tools.
- Understanding the spatial distribution of carbon sequestration potential can help prioritize areas for ecosystem-based mitigation approaches.

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Materials and Methods

Datasets. We calculated the carbon sequestration based on NEP (or NEE) and NBP outputs from two model intercomparison products and three data-driven products.

Model intercomparison products. We used the results of GPP, r_a, r_h, and NBP from a set of Earth System Models (ESMs) participating in the most recent climate-carbon simulations under the sixth phase of the Coupled Model Intercomparison Project (CMIP6) (36). Eight models provided monthly outputs for NBP, while eight (seven overlapping) supplied data for the remaining variables of interest in the esm-hist simulation (see Supplementary Table S1). These simulations span the period from 1850 to 2014 and are forced by prescribed anthropogenic CO_2 emissions rather than CO_2 concentrations. The data were obtained from The German Climate Computing Centre DKRZ. In addition, we used results from Dynamic Global Vegetation Models (DGVMs) compiled by the TRENDY v13 project (9). Twenty models provided monthly outputs of GPP, r_a, r_h, and 18 models provided monthly NBP (see Supplementary Table S2). We used experiment S3, which accounts for variations in atmospheric CO_2 and climate and spans the period from 1700 to 2023. Data were requested through the Global Carbon Budget.

Both CMIP6 and TRENDY intercomparison models were included in the analysis due to their distinct focuses. TRENDY DGVMs focus on simulating terrestrial vegetation dynamics and ecosystem processes, while CMIP6 ESMs provide a representation of the entire Earth system, incorporating simplified versions of DGVMs as subcomponents within a broader framework that also accounts for atmospheric, oceanic, and cryospheric processes (37, 38). For both CMIP6 and TRENDY models, NEP values are calculated as $NEP = GPP - (r_a + r_h)$.

Data-driven products. Besides the model intercomparison products, we utilized data from two atmospheric CO_2 inversion products: CarboScope and CAMS. These products estimate land-atmosphere net carbon exchange fluxes by combining CO_2 concentration measurements, transport models, and information on anthropogenic carbon emission and land-atmosphere carbon exchange (39). Specifically, we used the nbetEXToc_v2024E version of CarboScope (40), which provides daily data from 1957 to 2023 based on measurements from 173 sites. For CAMS (41), we used version v24, which includes monthly data from 1979 to 2020 derived from over 100 surface measurements sites. Additionally, we incorporate Net Ecosystem Exchange (NEE) from X-BASE produced with FLUXCOM-X (23), a machine-learning-based product trained on in-situ eddy covariance data. FLUXCOM-X provides monthly NEE estimates for the period 2001–2021

In this study, we assume that -NEE and NEP are interchangeable since both represent the same underlying processes—GPP minus respiration—despite being calculated using different data sources. Note the NEE is negative when the ecosystem assimilates more CO_2 than it releases, which is opposite to the convention for NEP.

Homogenization and ensemble calculations. To enable comparison, data from all products, as well as outputs from the different CMIP6 and TRENDY models, were regridded to a uniform $1° \times 1°$ spatial resolution using the bilinear interpolation method implemented via xESMF (42). We standardized the temporal resolution to monthly for all datasets that were not originally in monthly format. CMIP6 calculations utilized the full-time series (1850–2014), while comparisons across products were restricted to their overlapping period, i.e., 2001–2014. Positive NEP and NBP values were defined as fluxes moving from the atmosphere to the land, which contrasts with the conventions used in inverse models.

Then, to calculate the ensemble of CMIP6 and TRENDY models, we used a weighted averaging approach rather than a simple equal-weight average. This procedure assigns weights to each model based on its predictive performance against available data, following the multimodel inference framework presented by Burnham & Anderson (43), which builds on concepts developed by H. Akaike in the 1970s. The weighting procedure calculates each model's weight by measuring its distance from observed data. However, since we are dealing with global spatial models where direct observations are not available, we used data-driven products derived from sparse observations and scaled to the terrestrial surface. Specifically, we used FLUXCOM-X to calculate the NEP ensembles and CarboScope for NBP ensembles (chosen for its longer time series compared to CAMS). The calculation of weights (w_i) for each model i is detailed in (cite text with method). Once w_i are determined, the model ensemble $(\overline{x})$ is computed as

$$\overline{x} = \sum_{i=1}^{k} w_i \cdot x_i \qquad [1]$$

where k is the total number of models, and x_i are the values of the variable of interest for model i. In this case, x corresponds to NEP or NBP. The same weighting procedure is applied to create the ensemble of models from CMIP6 and TRENDY, using all data from periods that overlap with the available data periods of CarboScope (CMIP6:1957-2014, TRENDY:1957-2023) or FLUXCOM-X (CMIP6:2001-2014, TRENDY:2001-2021) (see Supplementary Figs. 1-4).

Carbon sequestration. After harmonizing all NEP (or NEE) and NBP data to the same temporal and spatial scale and calculating the ensembles for CMIP6 and TRENDY, carbon sequestration (CS) is computed. Here, CS is defined as the storage of a certain mass of carbon (M) within a system over a given period (t_0+T). It is quantified as the area under the curve representing the evolution of carbon mass over time (12), as

$$\mathrm{CS}\,(t_0, T) := \int_{t_0}^{t_0+T} \int_{t_0}^{t} s\,(t) - r\,(t)\, dtdt. \qquad [2]$$

where $s(t)$ is the gross flux rate of carbon removed from the atmosphere and stored into a system at a given time t, while $r(t)$ is the gross flux rate of carbon released from the system back to the atmosphere. This definition of CS accounts for both the fate of new carbon inputs and the legacy carbon already present in the system at t_0 (12). It considers the total carbon mass under evaluation as the net balance between all inputs and outputs over the specified time horizon (see Fig. 1 in Muñoz et al. (44)).

When calculating global carbon sequestration or aggregating it by latitudinal zones, the grid areas are factored into the computation, removing spatial units. Latitudinal zones are defined as tropical: 22.5°S-22.5°N, mid-latitudes: 22.5°S-55.0°S and 22.5°N-55.0S°N, and high latitudes: >55.0°N.

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